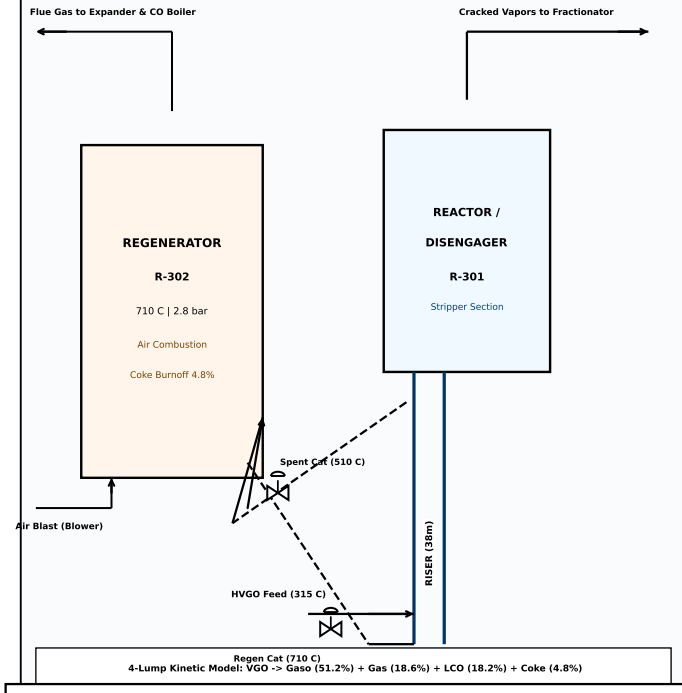
HEAVY-END UPGRADING & CONVERSION FLOWSHEETS: FCCU, HCU & DELAYED COKER

DETAILED CATALYTIC & THERMAL CRACKING ARCHITECTURES | RISER KINETICS, HYDROGEN BALANCE & DRUM CYCLES

FLUID CATALYTIC CRACKING UNIT (FCCU - U-300)

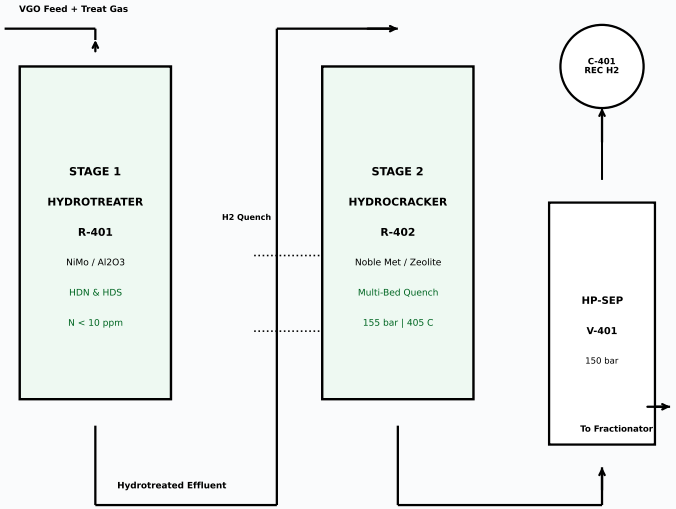
Feed: 30,000 BPD HVGO/LVGO | Catalyst: USY Zeolite | Riser: 535 C, 2.5 bar



4-Lump Kinetic Model: VGO -> Gaso (51.2%) + Gas (18.6%) + LCO (18.2%) + Coke (4.8%)

TWO-STAGE HYDROCRACKER COMPLEX (HCU - U-400)

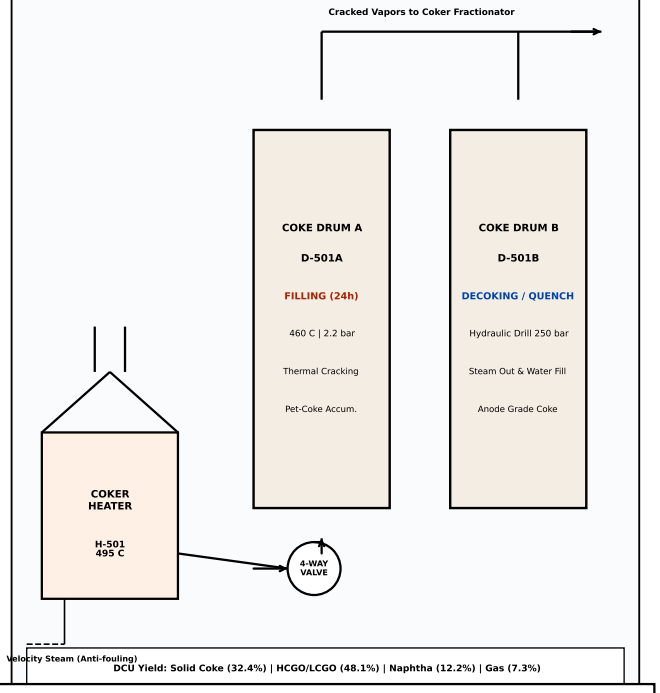
Feed: 25,000 BPD HVGO/HCGO | Pressure: 155 bar | Temp: 390-410 C | H2 Cons: 2.4 wt%



Yields: ULSD (48.4%, Cetane 58) | Kero/Jet (24.2%) | Naphtha (18.1%) | UCO Recycle (9.3%)

DELAYED COKING COMPLEX (DCU - U-500)

Feed: 18,000 BPD Vacuum Residue | Thermal Cracking: 495 C | Drum Cycle: 24h



DCU Yield: Solid Coke (32.4%) | HCGO/LCGO (48.1%) | Naphtha (12.2%) | Gas (7.3%)

REFINERY HEAVY-END UPGRADING UNITS: KINETIC, THERMODYNAMIC & OPERATIONAL MATRIX

CONVERSION UNIT / METRIC	FLUID CATALYTIC CRACKING (FCCU)	HYDROCRACKER COMPLEX (HCU)	DELAYED COKING UNIT (DCU)
Primary Objective & Function	Maximum High-Octane Gasoline (RON 93)	Max Middle Distillates (Jet/ULSD)	Zero Fuel Oil / Solid Carbon Reject
Operating Mechanism & Phase	Heterogeneous Fluidized Catalytic (Vap)	Fixed Bed Catalytic (Trickle Phase)	Thermal Non-Catalytic Pyrolysis (Liq/Vap)
Operating Pressure & Temp	2.5 bar(g) | 535 C Riser / 710 C Regen	155 bar(g) | 390 - 410 C Reactor Bed	2.2 bar(g) | 495 C Heater / 460 C Drum
Catalyst System & Form	Zeolite Y / USY Microspheres (75 um)	NiMo (Stg 1) + Pt/Zeolite (Stg 2)	None (Free Radical Thermal Polymerization)
Feedstock Quality & Diet	HVGO + LVGO (Conradson Carbon < 0.5%)	HVGO + Heavy Coker Gas Oil (HCGO)	Vacuum Residue (VR, 538 C+, CCR 18-22 wt%)
Major High-Value Products	Gasoline (51.2%), Propylene/LPG (18.6%)	ULSD (48.4%, S<10ppm), Jet Fuel (24.2%)	LCGO/HCGO (48.1%), Anode Coke (32.4%)
Hydrogen Consumption / Offgas	Negligible H2 (+0.2% Net offgas H2)	High Cons: 2.4 - 2.8 wt% Feed H2	None (Net Offgas Rich in C1-C4 Fuel Gas)
Decoking / Regeneration Mode	Continuous Air Blast in R-302 (CO2/CO)	Cyclic in-situ Catalyst Regeneration (2-3 yr)	24h Batch Hydraulic Drill (Water 250 bar)
Downstream Product Quality	RON 92.5 - 94.0 | Olefinic Gasoline	Cetane Index > 58 | Sulfur < 8 ppm	Coker Naphtha (Needs HDS) | Sponge Coke
Gazebo & ROS 2 Simulation Tag	virtual_refinery/fccu_node (Kinetics)	virtual_refinery/hcu_node (155 bar)	virtual_refinery/coker_node (Drum A/B)

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS PLANT DIGITAL TWIN SIMULATION

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
FCCU, HYDROCRACKER & DELAYED COKER CONVERSION FLOWSHEET	PROCESS PLANT DIGITAL TWIN & CAD	B.E. CHEMICAL ENGINEERING	DIAGRAMMATIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/VP/DWG-03	19 / 10 / 2026

VIRTUAL PETROLEUM REFINERY | 100,000 BPD CRUDE DIET | GAZEBO & ROS 2 DIGITAL TWIN | API / ASME BPVC